Paper Selection Criteria

Sep 24, 2026 | Author: Melanie Kurimchak | Contributors: Person



The GenAI Evidence Hub is a structured literature review of recent research into using Generative AI in automated scoring, item generation, and formative feedback. For more information about the study, see the pre-registration at https://osf.io/ar3yg/overview.

This document provides the plain-text version of the triageQ criteria profile developed during the initial stages of the project and included in the repository as an example (genai-evidence-hub-profile.json).

Criteria 1: GenAI Usage

Eligible papers must describe research published in 2023 or later that uses a generative AI model as the primary AI system.

Include if:

- The research employs an LLM or generative model (GPT-3/4/4o, Claude, Gemini, LLaMA, Mistral, DeepSeek, T5, BERT variants used generatively, etc.)
- The research uses an ensemble model that combines a generative AI component with traditional ML

Exclude if:

- The research uses only traditional or discriminative ML (SVM, Random Forest, KNN, logistic regression, XGBoost, CNN/RNN without a generative LLM component)
- GenAI is mentioned only in the literature review and not actually used in the study

Note: the study must have been conducted after 2020.

Part of the GenAI Evidence Hub, a Learning Data Insights project reviewing research on evaluation practices for generative AI in educational assessment. © 2026 Learning Data Insights. Licensed under CC BY 4.0.

Criteria 2: Relevant Assessment / Use Case Domain

Research must directly perform one or more of the domains below using GenAI. Discussing or mentioning a domain is not sufficient. The GenAI system must execute the task.

Item Generation
GenAI directly generates assessment questions, test items, prompts, or rubrics. Covers any item type: MCQ, short answer, essay prompts, simulation tasks.

- Not this domain if GenAI generates other content, such as stories or summaries, that is not used as an assessment item.

Formative Feedback
GenAI generates feedback text delivered to students to improve their learning. Feedback is tied to student work or responses, not just general content.

- Not this domain if the paper evaluates whether humans can detect AI-generated text, annotates data for a future feedback system, or generates scoring labels without student-facing feedback.

Automated Item Scoring
GenAI assigns scores, grades, or ratings to student-produced work (including synthetic work) that is typically scored by humans, such as essays, short constructed responses, code, drawings, or simulations. Includes holistic scoring, trait scoring, and rubric-based scoring.

- Not this domain if GenAI classifies, annotates, or labels text for an NLP/ML pipeline without producing a score on student work, if it detects whether text is AI-generated, or if it scores non-student content.

Multimodal Inferences
GenAI processes audio or video from classroom contexts to make assessment inferences, including speech recognition, behavioral coding, and engagement detection from A/V data.

- Not this domain if the paper uses text only, or if the multimodal data is not from a classroom or learning context.

Exclude if:

- The research task is AI detection or plagiarism detection
- The research task is data annotation or labeling for training future models

 © 2026 Learning Data Insights. Licensed under CC BY 4.0.

- The educational context is only background framing and not the actual study context
- The paper addresses fairness analysis only, with no assessment task

Tagging: record which domain(s) apply under Domains Identified. A paper can span more than one domain. Use "Unknown" only when you are certain the paper does not fit any of the four domains; it is not a hedge, and a reasonable case for a named domain should be tagged as that domain.

Criteria 3: Quality Assurance

Papers must report quantitative evidence evaluating the GenAI system's performance or impact. Either path below satisfies this criterion on its own; a paper does not need both.

Path A — Direct output evaluation (metrics that evaluate GenAI outputs against a standard)

- Precision, recall, F1-score (with baseline or comparison group)
- Accuracy compared with an established benchmark or human raters
- Cohen's Kappa, Weighted Kappa, Quadratic Weighted Kappa (QWK)
- AUROC, BLEU, ROUGE, GLEU, BERTScore
- Pearson or Spearman correlation with human scores
- Agreement rates (exact, adjacent) compared to human rater agreement
- Human rater evaluation of GenAI output quality (Likert ratings, satisfaction scores)

Path B — Outcome-based evidence (quantitative evidence of the GenAI system's real-world impact)

- Randomized controlled trials or quasi-experimental designs measuring student learning gains
- Pre/post comparisons of student performance attributable to the GenAI system
- Effect sizes (Cohen's d, partial eta-squared, etc.) from experiments comparing GenAI to a control
- Statistical tests (t-tests, ANOVA, regression) on learning outcomes where GenAI is the intervention
- Engagement or behavioral metrics tied to GenAI use in a learning context

Actual student learning outcomes are a valid and sufficient measure of GenAI system quality on their own. A well-designed RCT showing that AI-generated feedback improved student scores satisfies this criterion without also needing Path A metrics.

Exclude if:

 © 2026 Learning Data Insights. Licensed under CC BY 4.0.

- The paper reports only qualitative findings with no quantitative data
- The paper describes a system without any empirical evaluation
- All quantitative metrics are for a non-GenAI baseline, with no GenAI-specific results reported
- The paper is a literature review or theoretical framework without empirical results

Coding Sheet Fields

GenAI	Is GenAI used?
Domain	Does the research fit into one of the 4 domains (item generation, formative feedback, automatic scoring, multimodal inferences)?
Quality Assurance	Are the methods used to create outputs and evaluate results described in sufficient detail to identify them in our analysis?
Inclusion Decision	What is the reviewer's recommendation of the suitability of this study, given the research questions and inclusion criteria?
Inclusion Comments	Comments if needed to further explain decision (especially if a weak response or a No)

Inclusion Decision scale and confidence mapping:

Decision	Meaning
Strong Yes	All three criteria are unambiguously met based on explicit statements in the paper. No boundary judgment was required.
Weak Yes	The paper meets all three criteria, but at least one required meaningful interpretation, such as a topic adjacent to but not clearly within scope, an intervention role that is secondary or unclear, or metrics reported for a proxy task rather than the main outcome.
Weak No	The paper fails at least one criterion, but that judgment required meaningful interpretation rather than an explicit statement in the paper.

 © 2026 Learning Data Insights. Licensed under CC BY 4.0.

Strong No	At least one criterion is unambiguously not met based on explicit statements in the paper. No boundary judgment was required.
Uncertain	The paper sits on a genuine boundary. Key information is missing or contradictory, and a reasonable reviewer could interpret the study either way.

Inclusion Criteria (quick reference)

Criteria	Examples/Conditions
Must use generative AI	Such as GPT-4o, Claude Sonnet 4.5. Ensemble with an ML approach is okay too.
Must be an educational context	Tutoring is okay. Teaching content or a socioemotional skill is okay.
Must contain one or more of the four domains	Can be more than one domain within a paper.

Exclusion Criteria

Criteria	Examples/Conditions
Uses only traditional machine learning	K-Nearest Neighbors, Random Forest, etc., without being in an ensemble model with an LLM
Addresses only fairness	A fairness approach with no assessment task and no evaluation
Does not describe evaluation methods in sufficient detail	Not enough for us to make sense of what was done or to determine validity of outputs

 © 2026 Learning Data Insights. Licensed under CC BY 4.0.